

DEX-ROB LAB · GRADUATE RESEARCH STRATEGIC BRIEFING

AUTONOMOUS AGENT-IN-THE-LOOP REINFORCEMENT LEARNING:
LLM-GUIDED REWARD SYNTHESIS & POLICY OPTIMIZATION IN ISAAC
LAB

Methodological Framework for Bimanual Deformable Slicing (DOM) on Dual ARX AR5-L6 Arms with PhysX 5 FEM

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Lab & Dept: DEX-ROB Lab ·
SEAE, TJUTarget Platform: ARX AR5-L6 (7-DoF)
& Isaac Lab

EXECUTIVE SUMMARY & STRATEGIC MEETING INTENT

Reinforcement learning (RL) in continuous robotics control is fundamentally bottlenecked by the *reward engineering problem*: human researchers spend weeks manually shaping scalar reward weights to circumvent policy failures such as reward hacking, local minima, and force chatter. In this strategic briefing, we prepare our core proposal for Prof. Shan An: connecting a Large Language Model (LLM) reasoning engine directly with **NVIDIA Isaac Lab**. Building upon seminal frameworks like **Eureka (NVIDIA, ICLR 2024 Oral)** and **DrEureka [2024]**, the agent programmatically synthesizes vectorized reward functions in Python, executes massively parallel GPU rollouts, and refines both reward shaping and domain randomization bounds via in-context reflection. We investigate its decisive utility for our upcoming paper on **Bimanual Deformable Object Manipulation (DOM)** on our lab's dual **ARX AR5-L6 7-DoF arms** with LinkerHand O6 hands, where it systematically resolves the intractable "Tear-versus-Slip" trade-off in soft produce manipulation.

1. PROBLEM FORMULATION: THE BIMANUAL REWARD ENGINEERING BOTTLENECK

In a standard Markov Decision Process defined by $\langle S, A, P, R, \gamma \rangle$, the policy optimization objective is to discover parameters θ^* that maximize the expected discounted return:

$$J(\theta) = \mathbf{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right]$$

While sparse rewards (e.g., +1 on task completion) are theoretically unbiased, they yield intractable exploration complexity in high-dimensional continuous robot control. Consequently, roboticists rely on hand-crafted dense rewards composed of multi-objective weighted sums:

$$R_{\text{dense}}(s, a) = w_{\text{pen}} R_{\text{pen}} + w_{\text{saw}} R_{\text{saw}} - w_{\text{slam}} P_{\text{slam}} - w_{\text{damage}} P_{\text{damage}} - w_{\text{slip}} P_{\text{slip}} - w_{\text{reg}} \|a\|^2$$

In deformable and continuum manipulation, this conventional manual shaping suffers from three fatal shortcomings:

- **Post-Puncture Force Discontinuity:** In soft produce slicing, puncturing the outer peel into the flesh induces an abrupt reduction in cutting resistance (frequently dropping by 50%–75% within milliseconds, depending on blade geometry, approach velocity, ripeness, and cultivar). Linear penalty approximations fail to arrest downward knife momentum, causing the blade to slam into the interior and crush the food.
- **The Bimanual Pareto Conflict:** Ahmed's holding arm must maintain normal force strictly inside an experimentally calibrated compliance window ($F_{\text{min, slip}} < F_{\text{hold}} < F_{\text{max, damage}}$) to resist Shahd's sawing shear force (F_x) without inducing localized tissue bruising. Manually balancing multiple coupled scalar weights across two 7-DoF arms requires weeks of iterative retuning.
- **Proprioceptive Torque Sensing Constraint (Lab Hardware Confirmed):** As confirmed with Prof. Shan An, the physical laboratory setup operates with native joint torque sensing ($\tau_{\text{ext}} \rightarrow F_{\text{ext}}$ via Jacobian transpose) and camera vision, rather than specialized dense tactile skin arrays. Reward formulations must therefore operate robustly over proprioceptive torque signals and FEM state variables.

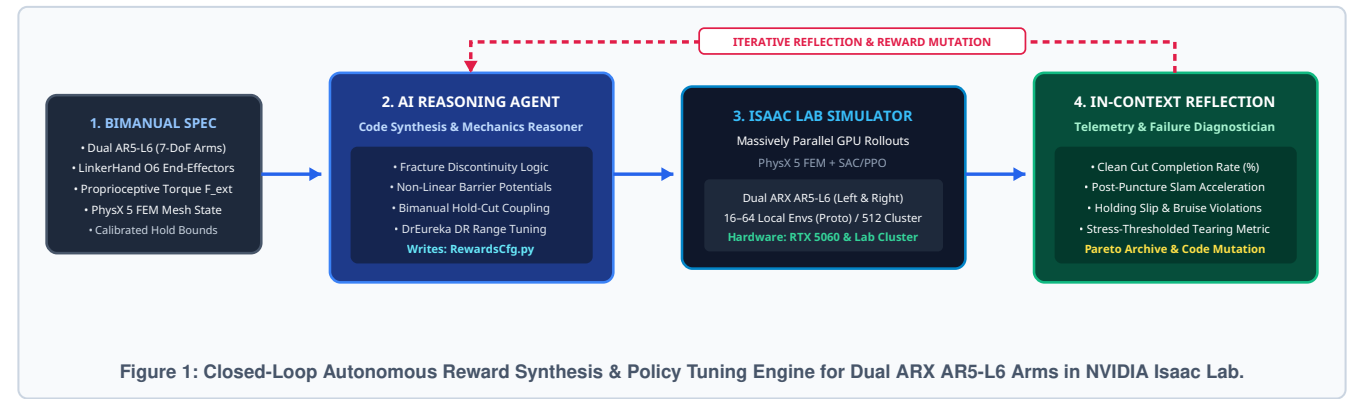
2. LITERATURE REVIEW: SOTA PRECEDENTS & FOUNDATION BENCHMARKS

The proposed methodology synthesizes key breakthroughs in LLM code generation and GPU-parallel robot learning. The table below outlines the core literature establishing this paradigm:

Framework / Paper	Authors & Institution	Core Algorithmic Innovation	Demonstrated Experimental Results
Eureka [1]	Y. J. Ma et al. <i>NVIDIA, Stanford, Penn</i> ICLR 2024 (Oral)	Zero-shot LLM reward code generation via evolutionary search with in-context reflection over raw Isaac Gym simulation logs.	Outperformed human expert rewards on 83% of 29 benchmarks across 10 robot embodiments; achieved 52% average normalized improvement across the suite.
DrEureka [2]	Y. J. Ma et al. <i>NVIDIA & Penn</i> arXiv 2024	Extends Eureka to sim-to-real transfer. Simultaneously discovers reward code and domain randomization (DR) ranges (friction, damping, latency).	First zero-shot sim-to-real transfer of quadruped locomotion atop rolling yoga balls and varied physical terrain without real-world tuning.
IsaacLabEureka [3]	NVIDIA Corporation <i>Official Framework</i> GitHub 2024	Isaac Lab implementation of Eureka. Public implementation is based on <code>DirectRLEnv</code> with RSL-RL and RL-Games backends, with OpenAI/Azure API support.	Adapting the framework to our custom bimanual deformable task and alternative LLM backends (Anthropic Claude, local models) represents a key engineering contribution.
Text2Reward [4]	T. Xie, S. Zhao, C. H. Wu et al. <i>ICLR 2024</i>	Generates dense reward code from natural language instructions with compiler verification and user preference feedback.	Demonstrated zero-shot generation across 17 manipulation tasks on ManiSkill2 and MetaWorld, and 4 MuJoCo locomotion environments.
DexCatch & Open TeleDex [5, 6]	F. Lan et al. [5]; S. An et al. [6] <i>CoRL 2024; DEX-ROB Lab</i>	[5] Dynamic dexterous catching baseline. [6] Lab-developed teleoperation platform establishing dual-arm kinodynamics and teleoperation infrastructure.	Provides laboratory-validated kinodynamic baseline and trajectory data for dual-arm robotic manipulation at Tianjin University.
AutoRL & Orbit [7, 9]	Chiang et al. [7]; Mittal et al. [9] <i>IEEE RA-L 2019, 2023</i>	[7] Evolutionary reward search in continuous control. [9] Unified simulation framework for robot learning in Omniverse, foundation of Isaac Lab.	Demonstrates the superiority of automated reward optimization and GPU-accelerated parallel simulation for complex manipulation.

3. PROPOSED CLOSED-LOOP SYSTEM ARCHITECTURE FOR ISAAC LAB

We propose deploying an autonomous orchestrator script (`agent_tuner.py`) running locally that interfaces directly with Isaac Lab. The closed-loop architecture coordinates four tightly integrated engines:



3.1 The 4-Stage Autonomous Pipeline

- Environment Serialization:** The agent reads the environment schema: dual ARX AR5-L6 7-DoF joint limits, LinkerHand O6 actuator states, Franka/ARX torque-reconstructed contact wrench, and PhysX FEM mesh nodal states.
- Evolutionary Code Synthesis:** The LLM generates K candidate reward function classes using Isaac Lab's native `RewTerm` decorators, implementing non-linear barrier potential mathematics.
- Massively Parallel Execution:** Isaac Lab runs headless vectorized rollouts across parallel GPU environments (16–64 local envs; 256–512 cluster envs) for 1,000 policy iterations.
- Telemetry Extraction & Reflection:** The log parser evaluates policy performance curves (reward progression, task success, violation count) and prompts the LLM with concrete diagnostic feedback to guide mutation for the next generation.

4. DIRECT APPLICATION: DUAL ARX AR5-L6 DEFORMABLE SLICING (DOM)

In our research paper (Paper 1), two collaborative **ARX AR5-L6 7-DoF arms** (actuated joints 1 to 7, equipped with **LinkerHand O6** 5-finger dexterous hands, defined in our stack via `AR5_L6_left.usda`, `AR5_L6_right.usda`, and `ar5_o6_left_combined.usda`) must execute coordinated holding, tensioning, and cutting of deformable biological produce modeled with **PhysX 5 FEM** continuum mechanics.

Resolving the "Tear-versus-Slip" Paradox in Soft Tissue Manipulation

Soft tissue exhibits highly non-linear, rate-dependent stress-strain behavior. If Gripper 1 exerts insufficient grip force, the tissue slips under lateral sawing drag ($F_{\text{hold}} < F_{\text{min, slip}}$). If Gripper 1 or Cutting Arm 2 exerts excessive force, localized shear stress exceeds critical yield limits ($\sigma > \sigma_{\text{crit}}$), inducing premature mesh rupture and tissue crushing.

Why the AI Agent Prevails Over Human Engineering: Human researchers attempt linear combinations of penalties that destabilize training. The LLM agent autonomously synthesizes *non-linear soft barrier potentials* that remain near zero during normal operation and asymptotically spike to $-\infty$ as the FEM stress tensor approaches rupture limits.

```
# Synthesized Isaac Lab Reward Class for Bimanual ARX AR5-L6 Slicing
@configclass
class BimanualTomatoCuttingRewardsCfg:
    # 1. Primary Objective: Continuous blade advancement along the cut trajectory
    cutting_progress = RewTerm(func=mdp.cutting_path_progress, weight=6.0)

    # 2. Sawing Motion Reward: Longitudinal velocity reduces normal penetration resistance
    sawing_efficiency = RewTerm(
        func=custom_sawing_shear_reward,
        weight=3.0,
        params={"min_sawing_vel_m_s": 0.02, "contact_force_threshold_N": 0.5}
    )

    # 3. Post-Puncture Slam Penalty: Exponential penalty on vertical blade acceleration
    post_puncture_slam_barrier = RewTerm(
        func=mdp.exponential_acceleration_barrier,
        weight=-10.0,
        params={"acceleration_threshold_m_s2": 1.5, "force_drop_ratio": 0.5}
    )

    # 4. Holding Arm Normal Force Compliance: Calibrated safe holding window
    holding_force_window = RewTerm(
        func=custom_holding_compliance_barrier,
        weight=-8.0,
        params={"f_min_slip_N": 1.5, "f_max_damage_N": 5.0} # Calibrated per cultivar
    )

    # 5. Action Regularization: Suppresses motor torque chatter on delicate tissue
    action_rate_l2 = RewTerm(func=mdp.action_rate_l2, weight=-0.005)
```

5. COMPUTATIONAL BUDGET & WORKSTATION FEASIBILITY ANALYSIS

Isaac Sim officially specifies a minimum requirement of 32 GB RAM and 16 GB VRAM. To ensure operational feasibility within DEX-ROB Lab, we implement a **two-tier execution strategy** that leverages our local prototyping workstation alongside the laboratory GPU compute cluster:

Resource Dimension	Tier 1: Local Prototyping Testbed	Tier 2: Lab Cluster Scaling	Operational Role & Benchmarking Status
GPU VRAM	NVIDIA RTX 5060 Laptop (8,151 MB GDDR7)	NVIDIA RTX 4090 / A6000 (24–48 GB VRAM)	Local: 16–64 envs for syntax verification and short rollouts. Cluster: 256–512 envs for full training.
Host System RAM	16 GB DDR5 5600 MHz	64 GB–128 GB Server DDR5	Headless execution minimizes memory footprint. High-env FEM rollouts scale to lab server.
Compute Cores	Intel Core i7-14650HX (16 Cores / 24 Threads)	AMD EPYC / Intel Xeon 32+ Cores	Enables fast compilation, log parsing, and local policy testing (~3–4 min per 1,000 steps).
LLM Inference Cost	API (Claude 3.5 Sonnet / GPT-4o)	API / Local DeepSeek-Coder	~1,500 tokens / generation; <\$2.00 total API cost for a 30-generation evolutionary run.

6. PROPOSED 4-PHASE RESEARCH ROADMAP (AHMED & SHAHD)

Estimated Target Implementation Schedule (subject to experimental FEM solver stability and hardware scaling):

Phase 1: Single-Arm Baselines (W1–W2) Construct isolated Isaac Lab stages: Shahd builds clamped cutting stage; Ahmed builds holding compliance using ARX AR5-L6 kinematics and torque force reconstruction.	Phase 2: Agent Harness Setup (W3–W4) Deploy lightweight Python orchestrator (<code>agent_tuner.py</code>). Connect LLM API to author candidate <code>RewardsCfg</code> files, run AST checks, and parse event metrics.
Phase 3: Bimanual Integration (W5–W6) Merge subsystems into full dual-arm <code>BimanualTomatoCuttingEnv</code> . Execute autonomous 20-generation evolutionary tuning runs to co-optimize holding and cutting feed.	Phase 4: Paper Finalization (W7–W8) Synthesize comparative ablation studies (Human baseline vs. Agent-synthesized reward vs. DrEureka domain randomization). Finalize manuscript for submission to IEEE RA-L / IROS.

7. STRUCTURED DISCUSSION AGENDA FOR MEETING WITH PROF. SHAN AN

KEY STRATEGIC QUESTIONS TO DISCUSS WITH ADVISOR Core Paper Positioning: Should we position this primarily as a <i>Deformable Slicing Paper with automated LLM tuning</i>, or as an <i>Autonomous Agent Framework for Complex Contact Physics</i>? (The former is highly favored in top robotics venues like IEEE RA-L, ICRA, and IROS). Baseline Comparison Matrix: We recommend benchmarking our method against three baselines: (a) classical human-engineered reward, (b) standard Cartesian impedance controller via MoveIt 2, and (c) heuristic trial-and-error RL. Sim-to-Real Testing Scope: Discuss whether physical dual-arm ARX AR5-L6 robot validation in the laboratory is mandatory for the initial manuscript, or if validated Isaac Lab PhysX FEM simulation with DrEureka domain randomization is sufficient for the first submission.
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8. ACADEMIC REFERENCES (IEEE FORMAT)

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[10] NVIDIA Corporation, "PhysX 5 Finite Element Method (FEM) Simulation for Deformable Solids and Cloth," *NVIDIA Omniverse Technical Documentation*, 2024.

9. ANTICIPATED SUPERVISOR QUESTIONS & DEFENSE STRATEGY

To ensure seamless communication and rigorous academic alignment during our supervisory meeting with Prof. Shan An, the table below maps critical questions to structured technical defenses grounded in first principles:

Anticipated Question	Underlying Academic Concern	Recommended Defense / Response Strategy
"Is the LLM fast enough to control the dual AR5-L6 arms in real time?"	Concern that LLM inference latency (500–1,500 ms) will bottleneck real-time robot control (50–100 Hz).	Separation of Timescales: The LLM is <i>not</i> in the low-level action loop. The RL policy (SAC/PPO) executes continuous torque control at 60 Hz natively on CUDA tensors. The LLM operates at the <i>meta-level</i> , synthesizing reward code once every 1,000 training iterations to analyze logs and refine objective code.
"Why not use Bayesian Optimization or Hyperband for reward weights?"	Concern that LLMs are an over-complicated substitute for standard black-box hyperparameter tuning.	Symbolic Formulation vs. Scalar Tuning: BO and Hyperband optimize scalar weights within a predefined parametric family. They cannot author symbolic logic, non-linear barrier potentials (e.g., $\exp(k(\sigma - \sigma_{\text{crit}}))$), or novel geometric invariants. LLMs reason across continuum mechanics and state semantics.
"Can our workstation handle 512 parallel Isaac Sim FEM environments?"	Concern that local hardware (RTX 5060 Laptop, 16 GB RAM) is below NVIDIA's 32 GB RAM / 16 GB VRAM minimum.	Two-Tier Strategy: We explicitly use the laptop as a Tier-1 local prototyping testbed with scaled environment counts (16–64 envs) for rapid code verification, syntax checking, and pipeline testing. Full-scale 512-env FEM training will scale to the DEX-ROB Lab compute cluster (RTX 4090 / A6000).
"How does the simulator handle soft body tearing without dynamic remeshing?"	Concern that PhysX 5 FEM does not provide out-of-the-box topological fracture or cutting sensors.	Stress-Thresholded Failure Metrics: PhysX 5 exposes nodal Cauchy stresses and displacements via GPU tensors. Rather than relying on unstable dynamic topological remeshing during initial training, our framework formulates a stress-thresholded penalty and rupture termination condition ($\sigma_{\text{vonMises}} > \sigma_{\text{crit}}$).
"How do you prevent the LLM from writing buggy code or crashing Isaac Lab?"	Concern that generated Python code will produce syntax errors, runtime exceptions, or CUDA memory leaks.	Compiler & AST Feedback Loop: The harness runs Python AST validation and a 10-step dry-run rollout before committing to full training. If an exception occurs, the traceback is reflected back to the LLM for immediate zero-shot code self-repair.

MEETING TAKEAWAY & IMMEDIATE NEXT ACTION

By demonstrating that the LLM agent operates strictly at the meta-level over symbolic formulation space, while hardware execution is safely partitioned across local prototyping and cluster rollouts, we present a scientifically mature, risk-mitigated research plan that directly advances DEX-ROB Lab's core research agenda in continuum manipulation.